

Commenters expressed the concern that wider spreads and reduced depth would negatively impact capital formation for growth companies.¹⁷⁹¹ Commenters specifically mentioned the importance of rebates for small and medium-sized growth companies, without which “market makers may no longer find it profitable to make tight markets.”¹⁷⁹² Two considerations enter in determining the effect of capital formation. First, for illiquid stocks, spreads are the primary determinant of revenue for liquidity providers. The rebate makes less of a difference on a percentage basis than for stocks that are more liquid. Second, the Commission does not expect the cost of transacting in illiquid securities to rise, net of fees and rebates.¹⁷⁹³ While the Commission acknowledges the crucial role of the ability of investors to transact for capital formation, it is not quoted spreads that matter to investors but rather the net spread available on exchange. In sum, liquidity is expected to improve for stocks with narrow quoted spreads that receive the tick size reduction—as discussed in section VII.D.1—and liquidity is not expected to be harmed for stocks that do not receive the tick size reduction—as discussed in section VII.D.2. Therefore, the amendments are expected to improve liquidity and thus will not impede capital formation through this channel.

F. Reasonable Alternatives

This section considers alternatives to the amendments. In the Proposing Release, we considered the benefits and costs of multiple categories of alternative, and variations within those

¹⁷⁹¹ See Nasdaq Letter I at 24.

¹⁷⁹² See Nasdaq Letter I at 25. See also Virtu Letter II at 10.

¹⁷⁹³ One commenter, stating that the reduced access fee cap would reduce “incentives for liquidity in thinly traded securities,” cited a study showing that an improvement in liquidity from stock splits resulted in significant reductions in the cost of capital for firms that did a stock split (Virtu Letter II at 8, citing Ji-Chai Lin, Ajai K. Singh & Wen Yu, Stock Splits, Trading Continuity, and the Cost of Equity Capital, 93 J. FIN. ECON. 474, 475 (Jan. 1, 2009)). As stated above, because the reduction of the access fee will not result in an increase in the cost of liquidity, see supra section VII.D.2.c, discussing this point, there is no reason to expect the cost of capital to increase as a result of lowering the access fee cap.

categories.¹⁷⁹⁴ For brevity we do not repeat that discussion here. Instead, this section focuses on additional alternatives suggested by commenters, to the extent they are not incorporated into the adopted amendments. We organize subsections around key elements of the Rule: tick size, minimum trading increment, access fee, and MDI and BOLO.¹⁷⁹⁵

1. Tick Size Alternatives

a. Alternative Criteria for Selecting Stocks Receiving a Smaller Tick Size

Commenters suggested alternative methodologies for identifying which stocks should receive a smaller tick size.¹⁷⁹⁶ These alternative methodologies are discussed in greater detail in section VII.D.1.b.iii and generally center on adding additional criteria, in addition to the TWAQS, to determine which stocks should qualify for a lower tick size.¹⁷⁹⁷ In that section, analysis failed to find evidence that the additional criteria would avert harm to market quality. One reason for this is likely that much of the information contained in these additional thresholds suggested by commenters is already contained in the TWAQS.¹⁷⁹⁸

Moreover, implementing these alternatives would increase the complexity of the amendments from the perspective of the listing exchanges, who would be required to track and implement multiple thresholds to identify tick-constrained securities. Increased complexity would increase the compliance costs of the amendments for these entities. Complexity would

¹⁷⁹⁴ See Proposing Release, supra note 11, at 80339.

¹⁷⁹⁵ Some commenters discussed “no action” as an alternative to the proposed rules. See, e.g., Virtu Letter II at 22-23. For purposes of the economic analysis, the baseline describes the world as it would exist without the rules.

¹⁷⁹⁶ See supra section VII.D.1.b.iii for additional discussion of these methodologies.

¹⁷⁹⁷ See supra note 1311 for discussion of specific commenter suggestions.

¹⁷⁹⁸ See supra note 1318 and surrounding discussion.

also increase for broker-dealers and investors, who would be required to take these changes into account. These alternatives would likely not affect the compliance costs of the rules for other market participants relative to the adopted amendments. This is because these alternatives would not change how these entities learn which stocks are subject to the \$0.005 tick and which are subject to the \$0.01 tick size in terms of assessing lists from the listing exchanges' websites, and the need to update systems to implement the different tick sizes.

The biggest effect of these alternatives relative to the adopted amendments is that they would reduce the number of securities receiving a reduced tick size. For example, one proposed alternative would limit the number of securities receiving a smaller tick size to an estimated 58 stocks.¹⁷⁹⁹ Commenters stated that limiting the sample via additional thresholds and criteria would ensure that only the stocks that are absolutely the most likely to benefit from a smaller tick size would receive the smaller tick size.¹⁸⁰⁰ However, the drawback to this more limited approach is that the analysis presented in sections VII.D.1.b.ii and VII.D.1.b.iii suggests that many stocks that would not qualify for the lower tick size under these alternative thresholds would likely still benefit from reducing the tick size. This conclusion is supported by the findings in Table 9 which demonstrate that across many dimensions TSP stocks with narrow spreads that are nonetheless in the bottom quartile based on depth, price, or trading volume, i.e., those that commenters suggest could perhaps be excluded from receiving the lower tick size, still experienced market quality improvements across many dimensions with a smaller tick. This analysis also fails to find statistically significant evidence that such stocks would be harmed. Consequently, adding additional criteria would add complexity to the implementation of the

¹⁷⁹⁹ See Cboe Letter II at 5.

¹⁸⁰⁰ See supra section VII.D.1.b.iii for additional discussion.

Rule, increasing the compliance costs of the rule, and would have lower benefits than the adopted amendments because it would leave some stocks with a wider tick size than would be optimal.

b. Alternative Threshold For Lower Tick Size

Some commenters suggested that the Commission adopt a threshold for the lower tick size that is different from the adopted amendments. The most common alternative suggested was a TWAQS of \$0.011 threshold.¹⁸⁰¹

The Commission estimates that the costs to implement this alternative would be similar to the adopted amendments because all affected entities would be required to perform the same work as in the adopted amendments. From an implementation perspective, the key difference between this alternative and the adopted amendments would be the considerably reduced number of stocks that would qualify for the alternative's lower tick size.

This alternative would more specifically target trading volume that is nearly always trading at the minimum trading increment. This alternative would leave stocks with quoted spreads between \$0.011 and \$0.015 with the \$0.01 tick size, whereas the adopted amendments assign a tick size of \$0.005 to such stocks. Using the same methodology as is used in Table 7 there would be an estimated 1,216 stocks receiving a \$0.005 tick size under this alternative, a reduction of approximately 572 stocks compared to the adopted amendments. These omitted stocks have between 1.1 and 1.5 ticks intra-spread on average. Research and Commission analysis as well as commenters' analyses suggest that 2 to 4 ticks intra-spread is likely an

¹⁸⁰¹ See, e.g., UBS Letter at 10 and JPMorgan Letter at 4. With a \$0.011 threshold, following the methodology employed in Table 7, an estimated 1,216 stocks would receive the lower tick size, with a \$0.02 threshold an estimated 2,339 stocks would receive the lower tick size.

optimal range for stocks.¹⁸⁰² Thus, assigning stocks with a quoted spread between \$0.011 and \$0.015 to a tick size of \$0.01 – resulting in 1.1-1.5 ticks intra-spread – is likely to produce worse market quality outcomes than assigning these stocks a \$0.005 tick – which would result in 2.2-3 ticks intra-spread. Thus, under this alternative these stocks, on average, would be expected to have lower overall market quality relative to the adopted amendments. Specifically, analysis in Table 8 of stocks in bin 2, which had 1-2 ticks intra-spread during the TSP, experienced significant improvements in market quality when the TSP tick size was relaxed – providing evidence that such stocks would benefit from a lower tick size. Consequently, this alternative, by failing to reduce the tick size for stocks that evidence suggests would benefit from a tick size decrease, would have lower benefits compared to the adopted amendments, while having similar costs.

c. Alternative Measurement Horizons for TWAQS and Effective Periods for Tick Sizes

Some commenters suggested alternative measurement horizons to determine the TWAQS as well as alternative periods that the tick size would be effective.¹⁸⁰³ Much of the analysis of alternative measurement horizons for the time weighted quoted spread as well as alternative operative periods for tick sizes is contained in VII.D.1.d. In sum, the analysis in that section presents tradeoffs. On the one hand, a shorter evaluation period ties the tick size to the most recent market experience for a given stock potentially resulting in the most relevant tick size to be assigned. On the other hand, if that time period is associated with transient spikes in quoted spreads, such as during the first quarter of 2020 coincident with the onset of the Covid-19

¹⁸⁰² See supra section VII.D.1.b.ii.

¹⁸⁰³ See, e.g., FIA PTG Letter II at 2 and JPMorgan Letter at 4.

pandemic, then the time period used to assign tick sizes would not be representative of current market conditions and a stock may be assigned a sub-optimal tick size.

There is also a tradeoff associated with the length of time that tick sizes are effective. More frequent updating means the tick size can adjust more rapidly to changes in the trading environment for a given stock and thus could increase the amount of trading volume associated with optimal tick sizes. The downside is that more frequent updates would also increase the cost and complexity of the amendments as market participants would have to adjust to tick sizes that change more frequently.

The Commission analyzed many iterations of evaluation period and tick size operative period and found evidence consistent with these tradeoffs.¹⁸⁰⁴ Consequently, depending on the combination of period used to determine the TWAQS and the effective period for the tick size, the total fraction of trading volume that trades in the preferred range may increase or decrease relative to the adopted amendments as suggested by the analysis in Table 10. Additionally, the costs and complexity of the alternatives would similarly be affected by alternative horizons chosen with more frequent updating associated with higher costs and complexity and less frequent updating associated with lower costs and complexity relative to the adopted amendments.

For example, as shown in Panel A of Table 10, an alternative which would have a one-month evaluation period and a one-month effective period for a tick size would reduce the amount of estimated trading volume that trades with a wide tick and likely would have benefited from a smaller tick to 8.5%, compared to an estimated 13.7% in the adopted amendments.

¹⁸⁰⁴ See supra section VII.D.1.d.

Consequently, more trading volume would be assigned a tick size that is expected to improve market quality for the stock. However, relative to the adopted amendments this alternative would have 12 tick size changes per year instead of the adopted 2 changes, thus it would increase the complexity and compliance costs associated with the rule. Overall, relative to the adopted amendments, this alternative results in a significant increase in complexity but only achieves a relatively modest increase in effectiveness in terms of trading volume with the appropriate tick size.

On the other side of the spectrum, a rule that uses a 12-month operative period could produce the opposite effect. It would reduce complexity somewhat by reducing the number of revisions, but with a significant decrease in effectiveness of the tiered tick size regime. For instance, again using data from Table 10, this alternative would increase the amount of trading volume that retains the larger tick size but would likely benefit from a smaller tick size to 20.8%, up from an estimated 13.7% associated with the adopted amendments.

2. Access Fee Alternatives

a. 15 mils / 30 mils Access Fee

Some commenters recommended adopting a two-tier approach to existing Rule 610(c)'s uniform maximum access fee cap. Specifically, these commenters recommended an access fee of 15 mils for stocks with a \$0.005 cent tick and maintaining the 30 mils maximum access fee for

stocks that continue to have a \$0.01 cent tick (15 mils/30 mils alternative).¹⁸⁰⁵ This 15 mils/30 mils alternative would be applied to stocks priced \$1.00 or more.¹⁸⁰⁶

According to one commenter, this approach would be beneficial because the higher rebate cap for \$0.01 tick stocks would maintain incentives to provide on-exchange lit liquidity.¹⁸⁰⁷ The Commission acknowledges that the lower access fee cap on stocks with the higher tick is likely to widen spreads. As discussed in sections VII.D.2.c and VII.E.1, these wider spreads will not lead to a diminution of lit liquidity on exchanges. Rather, liquidity providers will adjust their quotes to reflect the change in fees and rebates, resulting in higher quoted spreads without an increase in transaction costs nor a decrease in lit liquidity.

Relative to the adopted amendments, this alternative would raise costs by causing stocks to oscillate between the two tick sizes, resulting in the tick size being consistently mis-assigned for the oscillating stocks. Specifically, the 15 mils/30 mils alternative would result in a feedback loop scenario on tick size assignment from tying fee caps to tick size. As discussed in sections

¹⁸⁰⁵ See, e.g., NYSE Letter I at 7 and Nasdaq Letter IV at 2. Nasdaq's alternative assumes that the Commission will adopt only one additional reduced tick size bucket of \$0.005 and a uniform access fee of \$0.10. Nasdaq Letter IV. See also NYSE, Schwab, and Citadel Letter at 2 ("we recommend a reduction that is proportionate to the proposed reduction in the minimum quoting increment for tick-constrained symbols. This would reduce the current \$.0030/share cap to \$.0015/share for the symbols with a half-penny minimum quoting increment"); Nasdaq Letter I at 2; MMI Letter at 7 ("access fees should be scaled based on 30% of the minimum pricing increment"); Robinhood Letter at 5, 56-59 ("the Commission should tie access fee caps to be consistently proportional to the applicable tick size at the current proportion of 30%"); and MEMX Letter at 23-24 ("Lower the access fee cap in tick constrained NMS Stocks to \$0.0015 to maintain the proportionality of access fees and tick sizes, and include auction fees within the scope of the rule to prevent competitive distortions that would otherwise result if listing exchanges were permitted to use auction fees to avoid a lower fee cap").

¹⁸⁰⁶ Commenters recommending this alternative did not address the treatment of sub \$1.00 stocks. For purposes of this discussion, we assume a proportionately reduced 0.15% access fee cap for those stocks. This access fee cap percentage relative to the adopted amendments would mitigate the expected reduction in exchanges' revenue resulting the lower access fee cap. Cf. Cboe Letter I (not recommending any reduction in access fees but, if access fees are reduced, recommending that the access fee [for stocks priced equal to or greater than \$1.00] should not be reduced below \$.0015 for tick constrained securities with a \$.0005 [tick] increment. For securities priced less than \$1.00, the access fee cap must remain unchanged to support competition, differentiation, and liquidity provision.").

¹⁸⁰⁷ Nasdaq Letter IV at 11 and passim. See also supra notes 529 and 535 and accompanying text.

VII.B.3 and VII.C.2.b, access fees are tied to rebates, which in turn influence quoted spreads. Consider, for example, a stock that trades in an Evaluation Period, on average, with a 1.4 cent TWAQS and 30 mils access fee cap (and rebate). Under the amendments to Rule 612, this stock would receive a 0.5 cent tick. If this stock were to also to be subject to the 15 mils /30 mils alternative, it would thus be subject to a 15 mils fee cap once it receives the smaller tick size, instead of a 30 mils one, and its average TWAQS would increase to 1.7 cents (increase by twice the 0.15 cents reduction in rebates as result of the lower access fee cap).¹⁸⁰⁸ But with a TWAQS of 1.7 cents the stock in a subsequent Evaluation Period would receive a tick of 1 cent and the access fee cap would again be 30 mils.¹⁸⁰⁹ At that point, the aforementioned process would begin again: the stock would yo-yo between tick sizes of 0.5 cents and 1 cent and access fee caps of 15 mils and 30 mils, generating investor confusion and additional costs.¹⁸¹⁰ In contrast, consider what would happen to this same stock under a uniform 10 mils access fee cap. The average quoted spread would widen from 1.4 cents to 1.8 cents, at which point the stock would not be subject to a tick size reduction and there would be no oscillations. The same problem would occur even with a lower threshold for a tick size reduction (e.g., if the TWAQS threshold for the

¹⁸⁰⁸ It is not necessary to assume that the exchange rebates the entire access fee. Rather, it is sufficient that the difference between the rebate under the access fee cap of 0.30 mils and the rebate under the access fee cap of 0.15 mils is 0.15. The example is not changed if the rebate goes from 0.28 mils to 0.13 mils.

¹⁸⁰⁹ As discussed above in section VII.D.1, the stock's having a 0.5 cent tick would likely lead to a narrower quoted spread than 1.7 cents. However, if the smaller tick causes the quoted spread to fall by anything less than 0.2 cents, the tick and access fee cap would revert back to 1 cent and 30 mils after the next Evaluation Period.

¹⁸¹⁰ One commenter stated that, "the Commission should consider the possibility of tick size oscillation for some stocks that fall close to the threshold values for TWAS." The commenter stated that such oscillation will impose excessive costs. See Mitre Corp. Letter at 5. See also section VII.D.1.d for a quantitative analysis on the tradeoff between appropriate tick assignment and the number of tick changes when evaluating the length of the evaluation and operative periods. The oscillation discussed in this alternative does not present such a tradeoff—this oscillation results in both more tick changes and more tick misassignment.

0.5 cent tick were to be set at 1.1 cents instead of 1.5 cents, then the 15 mils/30 mils alternative would result in spreads oscillating between 1.05 and 1.35 cents for some stocks, causing them to yo-yo between tick sizes). Moreover, oscillation would also occur if one were to add other metrics to the TWAQS threshold for smaller tick sizes because there will still be stocks near the TWAQS threshold. The simplest, and perhaps the only, way to avoid this feedback loop is to use a uniform access fee cap.

The 15 mils/30 mils alternative would also increase complexity because the higher level of fees and rebates create a larger wedge between the quoted half-spread and the true cost of demanding liquidity.¹⁸¹¹ Further, this alternative would reduce some benefits related to the minimum pricing increment, most substantially for stocks with the penny tick. As discussed in section VII.C.1.b, these stocks may have, at times, an economic spread that is less than one penny. At those times, the difference in outcome between a 30 mils and 10 mils access fee would be a relatively large reduction in distortions.¹⁸¹² For stocks assigned to the half-penny tick that remain tick-constrained, the reductions would be more minor in an absolute sense (15 mils versus 10 mils). The analysis for potential conflicts of interest is similar.¹⁸¹³ While the Commission acknowledges that there is some access fee that would be so low as to create strain on exchange business models, 10 mils appears well above this point.¹⁸¹⁴ Like the adopted amendments, this alternative would not affect an exchange's ability to earn its baseline net

¹⁸¹¹ See supra section VII.D.2.d.

¹⁸¹² See supra section VII.D.2.d. For stocks receiving the penny tick, the reduction in the access fee cap from 30 mils to 10 mils will reduce transaction costs for stocks that experience periods in which they are tick-constrained, and further reduce the probability that a stock becomes tick-constrained.

¹⁸¹³ See supra section VII.D.2.d

¹⁸¹⁴ See supra section VII.D.2.b